Quantum Mechanics II — Exercise Sheet 2

Evolution and a position reading

Variant: Questions only

Exercise 1 (A state with a shorter revival). Let $\psi = \sum_{j\geq 0} c_{2j+1} u_{2j+1}$ be normalized. (i) Show that its physical state revives at $T_{\mathrm{rev}}/8$, where $T_{\mathrm{rev}} = 2\pi\hbar/E_1$. (ii) Decide whether this time is always the least positive revival time. For a state with both c_1 and c_3 nonzero and all other coefficients zero, find that least time.

Exercise 2 (A position filter at a node). Start with the box mode u_3 and retain a position reading in $(0, L/3)$. (i) Find its probability and conditional vector. (ii) Compute all sine coefficients of the conditional vector. (iii) Decide whether its mean energy is finite and whether it belongs to $D(H)$. (iv) Treat the coefficient at $n = 3$ separately.

Exercise 3 (Inferring a relative phase). The initial box vector is $\psi = (u_1 + e^{\mathrm{i}\phi} u_2)/\sqrt{2}$, with an unknown $\phi \in \mathbb{R}$. Position means are measured on separately prepared copies at $t = 0$ and $t = \tau = \pi\hbar/(6E_1)$. They are

$$\langle Q\rangle_0 = \frac{L}{2} + \frac{8L}{9\pi^2}, \qquad \langle Q\rangle_\tau = \frac{L}{2} - \frac{8\sqrt{3}L}{9\pi^2}.$$

Find ϕ modulo 2π and describe the ambiguity if only the first mean is known. Use the given matrix element $\langle u_1, Q u_2\rangle = -16L/(9\pi^2)$.

Exercise 4 (A coarse energy reading keeps coherence). Let $\psi = (u_1+u_2+u_3)/\sqrt{3}$. Compare a yes/no reading of energy in $\{E_1, E_2\}$, conditioned on yes, with a fully resolved energy reading whose result is forgotten after retaining only E_1 or E_2. Give the two conditional density operators and their later position means. Use $\langle u_1, Q u_2\rangle = -16L/(9\pi^2)$.